

Pure Bending of a Steel Beam

Strain distribution, neutral axis, and deflection of a simply supported beam under four-point loading

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Team	Kacper Czarnecki, Ashlee Hercules, Luis Ramirez, Jackson Robillard	My role	Individual discussion and conclusion; data analysis with team
Equipment	Four-point bending frame, electrical strain gauges, digital dial indicator		

PROJECT SUMMARY

This experiment tested classical beam theory on a simply supported steel beam in four-point bending, which produces a region of pure bending between the load points. Ten electrical strain gauges spaced through the depth of the beam recorded strain at loads of 2,000, 4,000, and 6,000 lb during loading and unloading, and a digital dial indicator measured mid-span deflection. Strain varied linearly with height at every load ($R^2 \geq 0.987$), confirming that plane sections remain plane, and linear regression located the neutral axis 4.13–4.47 in above the lowest gauge. Measured deflections matched theory with an average error of 1.9% (maximum 4.3%), and loading and unloading results nearly coincided, confirming elastic behavior.

1. Objectives & Theory

The goals were to locate the neutral axis, verify that plane sections remain plane during bending, and compare measured strains and deflections with values predicted by flexural theory.

In pure bending, beam theory predicts that strain varies linearly with distance from the neutral axis, with compression on one side and tension on the other. Deflection follows from the differential equation of the elastic curve, $d^2y/dx^2 = M/EI$, where M is the bending moment, E the elastic modulus, and I the moment of inertia. Integrating with the beam’s boundary conditions for four-point loading gives the deflection within the constant-moment region as $y = (Pa / 6EI)(3x^2 - 3Lx + a^2)$, where a is the distance from each support to its load point.

2. Results

Table 1. Measured strain ($\mu\text{in/in}$) by gauge and applied load P

Gauge	Elevation (in)	0 lb	2,000	4,000	6,000	4,000 (unload)	2,000 (unload)	0 (unload)
1	8.08	8	-194	-412	-628	-412	-182	6
2	6.69	0	-126	-282	-426	-260	-124	2
3	5.93	6	-80	-192	-296	-194	-84	7
4	5.12	10	-30	-118	-176	-148	-74	-1
5	4.50	4	-8	-42	-74	-42	-16	2
6	3.77	2	34	40	52	38	34	4
7	3.00	2	78	124	180	126	76	9
8	2.18	0	122	210	310	212	118	8
9	1.40	2	166	300	442	300	164	1
10	0.40	2	242	450	662	454	242	8
Deflection (in)		0	0.136	0.264	0.405	0.272	0.138	0

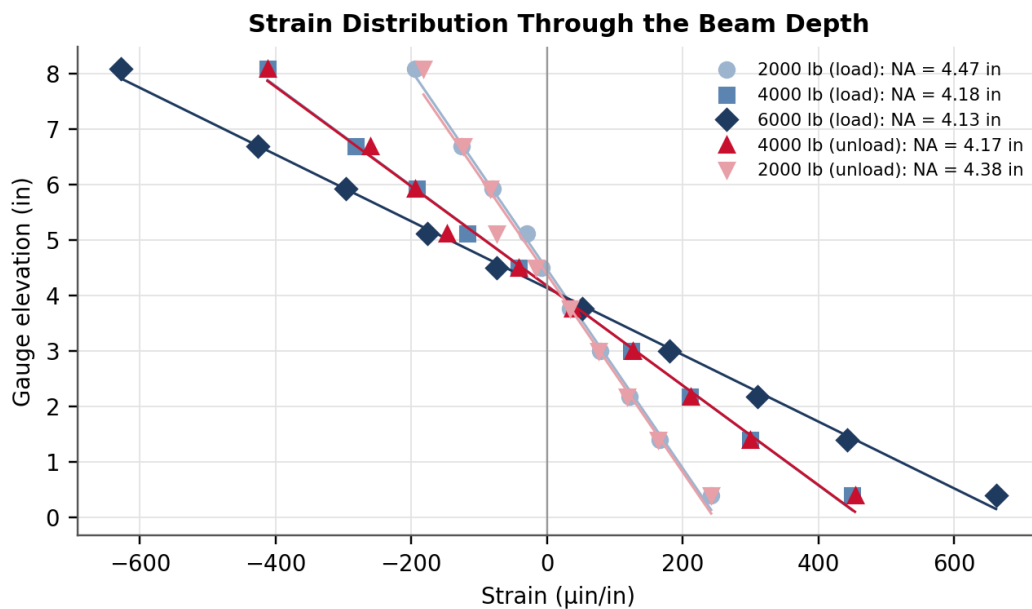


Figure 1. Strain versus gauge elevation at each load, with linear fits and the resulting neutral axis (NA) location.

Table 2. Theoretical vs. measured extreme-fiber strain (electrical gauges)

Load P (lb)	Theoretical ($\mu\text{in/in}$)	Bottom gauge (tension)	Error	Top gauge (compression)	Error
2,000 (load)	± 231	+242	4.8%	-194	16.0%
4,000 (load)	± 462	+450	2.6%	-412	10.8%
6,000 (load)	± 692	+662	4.3%	-628	9.2%
4,000 (unload)	± 462	+454	1.7%	-412	10.8%
2,000 (unload)	± 231	+242	4.8%	-182	21.2%

Table 3. Mid-span deflection

Load P (lb)	Theoretical (in)	Measured (in)	Error
2,000 (load)	0.138	0.136	1.4%
4,000 (load)	0.276	0.264	4.3%
6,000 (load)	0.414	0.405	2.2%
4,000 (unload)	0.276	0.272	1.4%
2,000 (unload)	0.138	0.138	0.0%
Average			1.9% (SD 1.4%)

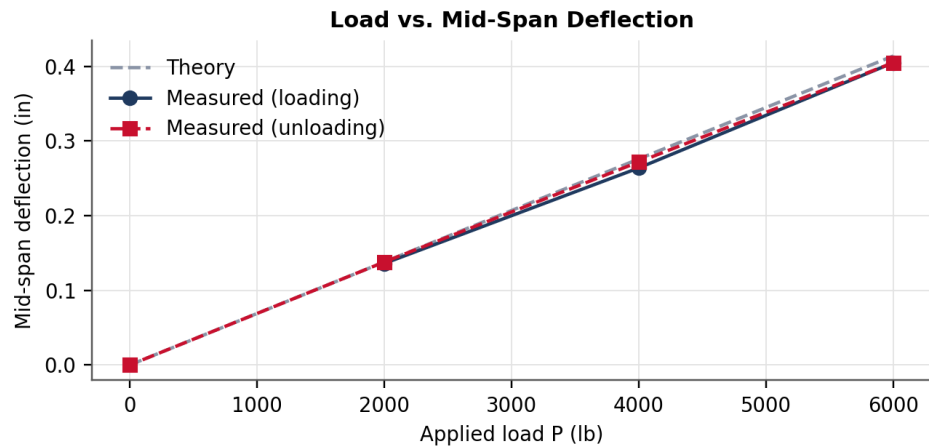


Figure 2. Measured mid-span deflection versus theory during loading and unloading.



Figure 3. Weights applied for the 2,000 lb load

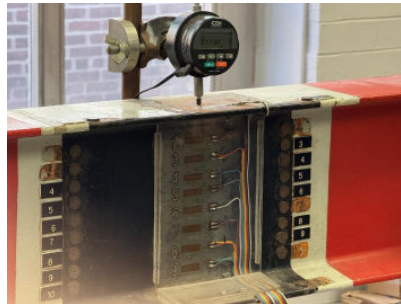


Figure 4. Strain gauges and deflection gauge



Figure 5. Full test setup

3. Discussion

Plane sections remain plane. At every load, strain varied linearly through the depth of the beam, with R^2 values of 0.987 to 0.998. This is the central assumption of flexural theory, and the data confirm it directly: strain grows in proportion to distance from the neutral axis, compressive above it and tensile below it.

Neutral axis. Linear regression placed the neutral axis between 4.13 and 4.47 in above the lowest gauge, with the three higher loads clustering tightly around 4.13–4.18 in. The 2,000 lb cases give slightly higher values because the strains are small, so gauge noise has a larger effect on the fit. The result is consistent with the data itself: gauges 5 and 6, closest to the neutral axis, recorded the smallest strains at every load.

Strain accuracy. The bottom (tension) gauge matched theory within about 5% at every load. The top (compression) gauge consistently read 9–21% below theory. Because the difference is one-sided and repeats at every load, it points to a systematic cause rather than random error, such as a gauge bonding or calibration difference, a slight asymmetry in the section, or local effects near the load points. It also nudges the fitted neutral axis slightly toward the compression side.

Deflection and elasticity. Measured deflections matched theory closely, averaging 1.9% error with a maximum of 4.3%. Loading and unloading values nearly coincided, and the beam returned to essentially zero deflection and strain when unloaded, so the beam stayed in the elastic range. The small differences between the two cycles, with slightly higher readings during unloading, may reflect minor settling, friction in the apparatus, or very slight inelastic behavior, but their effect on accuracy was minimal.

Sources of error. Likely sources include human error during setup and reading, oscillation of the apparatus during loading and unloading, small gauge placement errors, and natural variation in the material. None significantly compromised the results.

4. Conclusion

The experiment achieved every objective. Measured strains and deflections in the steel beam closely matched classical beam theory: strain varied linearly with height, confirming that plane sections remain plane; the neutral axis was located near its expected position; and deflections averaged under 2% error while returning fully on unloading. Overall, the results support beam theory as a reliable structural design tool, while also showing why real-world verification of theoretical models remains an important part of safe and effective engineering design.

References

- Ciesla, M. (2017). *Strength of materials laboratory manual*. New Jersey Institute of Technology.
- Beer, F. P., Johnston, E. R., Jr., DeWolf, J. T., & Mazurek, D. F. (2020). *Mechanics of materials* (8th ed.). McGraw Hill.